Metals Market Index MMI

MMi Dashboard

Iron Ore Price Indices

Iron Ore Port Stock (FOT Qingdao)
IOPI62 62% Fe Fines RMB/t

829
-15 -1.78%
Apr 28th, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPI65 65% Fe Fines RMB/t

940
-15.00 -1.57%
Apr 28th, 2023

Iron Port Stock (FOT Qingdao)
IOPI58 58% Fe Fines RMB/t

717
-1 -0.14%
Apr 28th, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI62 62% Fe Fines USD/dmt

140.50
0.00 0.00%
Apr 28th, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI65 65% Fe Fines USD/dmt

145.10
0.00 0.00%
Apr 28th, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPLI 62.5% Fe Lump RMB/t

945
-30 -3.08%
Week Ending Apr 28th, 2023

Exchange Traded Contracts

DCE Iron Ore 62% Fines
I2309 (Jan) RMB/t (3pm close)

714.00
0.50 0.07%
Apr 28th, 2023 (3pm close)

SGX Iron Ore (CFR Qingdao) 62% Fe Fines
May 23 USD/dmt

101.35
0.05 0.05%
Apr 28th, 2023 (5.30 pm Print)

SHFE Rebar
RB2310 (Jan) RMB/t

3660
-31 -0.84%
Apr 28th, 2023 (3pm close)

Freight Rates

C3, Tubarao - Qingdao USD/t

22.65
0.53 2.41%
Apr 27th, 2023

C5, W. Australia - Qingdao USD/t

9.10
0.56 6.56%
Apr 27th, 2023

Steel Price

Steel Rebar (China Domestic) RMB/t

3830
-140 -3.53%
Week Ending Apr 28th, 2023

Inventory Levels

Iron Ore Inventory at Chinese Ports (35)
million tonnes

125.16
-1.79 -1.41%
Week Ending Apr 28th, 2023

Steel Inventory in China
million tonnes

15.86
-0.54 -3.26%
Week Ending Apr 28th, 2023

Steel Price

Steel HRC (China Domestic) RMB/t

3980
-90 -2.21%
Week Ending Apr 28th, 2023

IRON ORE PORT STOCK INDEX (IOPI)

Apr 28th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOPI62	62% Fe Fines	829	-15	-1.8%	869	880	858	892	111.24	-2.18	-1.9%	120.06	120.58	117.93	122.93
IOPI58	58% Fe Fines	717	-1	-0.1%	813	798	761	793	96.74	-0.22	-0.2%	113.29	110.17	105.29	110.04
IOPI65	65% Fe Fines	940	-15	-1.6%	981	992	970	1003	126.76	-2.08	-1.6%	136.05	136.43	133.84	138.73

IRON ORE SEABORNE INDEX (IOSI)

Apr 28th, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOSI62	62% Fe Fines	140.50	0.00	0.00%	114.97	124.54	83.90	149.05
IOSI65	65% Fe Fines	145.10	0.00	0.00%	124.61	140.29	94.45	173.75

MARKET COMMENTARY

DCE iron ore futures fall today by 0.97%. the main contract I2309 closed 714. On the last day before Labor Day, some trade orders were closed and steel mills' pre holiday restocking was basically over. The overall transaction atmosphere in the market was relatively cold. PBF at Shandong port dealt 819-822 yuan/mt; increased 10-12 yuan/mt over yesterday. PBF at Tangshan port dealt 825-827 yuan/mt; increased 7-10 yuan/mt over yesterday. SSF at Tangshan port dealt 683 yuan/mt; increased 10 yuan/mt over yesterday. As of April 28th, the total inventory of 35 ports tracked by SMM was 125.16 million tons, a decrease of 1.79 million tons from last week and 9.95 million tons from the same period last year. The daily average port clearance volume of imported ore in this period increased by 51000 tons to 3.151 million tons on a weekly basis. Prior to the May Day holiday, the trading atmosphere in the iron ore market was active, driving the enthusiasm for port clearance to increase. This week's port clearance volume has risen to its highest level in six months. Currently, there are frequent reports of production reduction plans and blast furnace maintenance in domestic steel mills. In the future, the demand for finished products may be average, and the production of

IRON ORE PORT LUMP INDEX (IOPLI)

Week Ending Apr 28th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [3]						
Index	Fe Content	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]
IOPLI62	62.5% Fe Lump	945	-30	-3.1%	927	1030	770	1330	127.13	0.32	0.25%	120.05	137.80	95.00	180.00

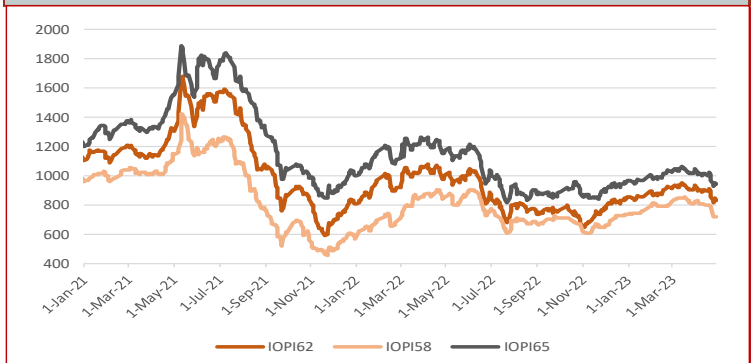
IRON ORE DOMESTIC CONCENTRATE SPOT PRICE ASSESSMENTS AND COMPOSITE INDEX

Week Ending Apr 28th, 2023				RMB/tonne (excluding tax) [3]				USD/tonne (excluding tax) [3]			
Province	Region	Product	Basis	This week	Change %	Low [2]	High [2]	This week	Change %	Low [2]	High [2]
Hebei	Hanxing	66% Fe Concentrate	Dry	1008	-1.1%	779	1645	146.53	-1.08%	110.31	255.69
Hebei	Qian'an	65% Fe Concentrate	Dry	985	-4.8%	780	1630	143.18	-4.83%	110.51	251.57
Liaoning	Anshan	65% Fe Concentrate	Wet	765	-3.8%	620	1310	111.20	-3.77%	87.40	202.32
Shandong	Zibo	65% Fe Concentrate	Dry	1022	-1.2%	800	1752	148.56	-1.16%	117.19	272.32

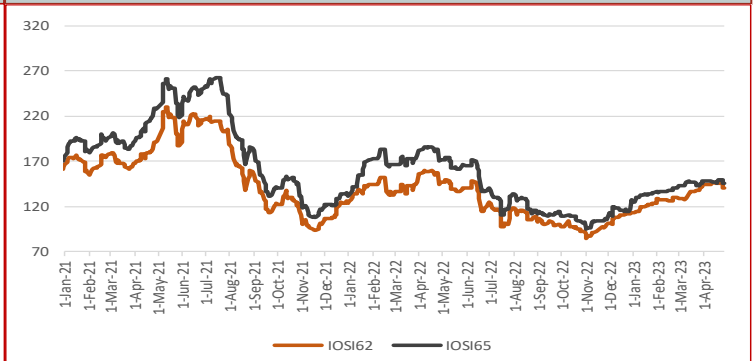
Week Ending Apr 28th, 2023	This week	Change %	Low [2]	High [2]
China Mines Concentrate Composite Index RMB/WT	888.41	-4.18%	706.36	1511.22

[1] Exchange rate applied: RMB/USD = 6.7522 [2] Last 12 months
[3] Weekly exchange rate applied: RMB/USD =6.87528

IRON ORE PORT INDEX, FOT QINGDAO (RMB/WT)



IRON ORE SEABORNE INDEX, CFR QINGDAO (USD/DMT)



IRON ORE PORT STOCK INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Apr 28th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne						
Index	Fe Content	December	January	February	March	MTD	QTD	YTD	December	January	February	March	MTD	QTD	YTD
IOPI62	62% Fe Fines	822	856	892	924	869	881	880	109.43	117.00	121.64	124.91	120.06	120.90	120.58
IOPI58	58% Fe Fines	711	751	799	832	813	792	798	95.17	103.34	109.79	113.41	113.29	109.35	110.17
IOPI65	65% Fe Fines	935	968	1003	1036	981	993	992	124.90	132.80	137.36	140.49	136.05	136.72	136.43

IRON ORE SEABORNE INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Apr 28th, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	December	January	February	March	MTD	QTD	YTD
IOSI62	62% Fe Fines	108.92	119.76	127.73	134.77	112.14	114.50	124.54
IOSI65	65% Fe Fines	116.83	132.39	138.06	145.14	122.86	124.61	140.29

FREIGHT RATES

Apr 27th, 2023		FREIGHT RATES - DRY BULK US$/wet tonne				
Route	Designation		Change	Change %	Low [2]	High [2]
W. Australia - Qingdao	C5	9.10	0.56	6.56%	3.57	16.77
Tubarao - Qingdao	C3	22.65	0.53	2.41%	6.70	36.40

IRON ORE PORT LUMP INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Week Ending Apr 28th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	December	January	February	March	MTD	QTD	YTD	December	January	February	March	MTD	QTD	YTD
IOPLI62	62.5% Fe Lump	922	964	988	1012	925	924	1032	117.06	126.47	129.95	131.15	119.39	120.08	137.99

IRON ORE INDEX PREMIUMS/DISCOUNTS

Apr 28th, 2023		PORT STOCK INDEX (RMB/WT)	
Index	Fe Content	Spread to IOPI62	% Spread to IOPI62
IOPI58	58% Fe Fines	-112	-13.51%
IOPI65	65% Fe Fines	111	13.39%

Apr 28th, 2023		SEABORNE INDEX (USD/DMT)	
Index	Fe Content	Spread to IOSI62	% Spread to IOSI62
IOSI65	65% Fe Fines	4.60	3.27%

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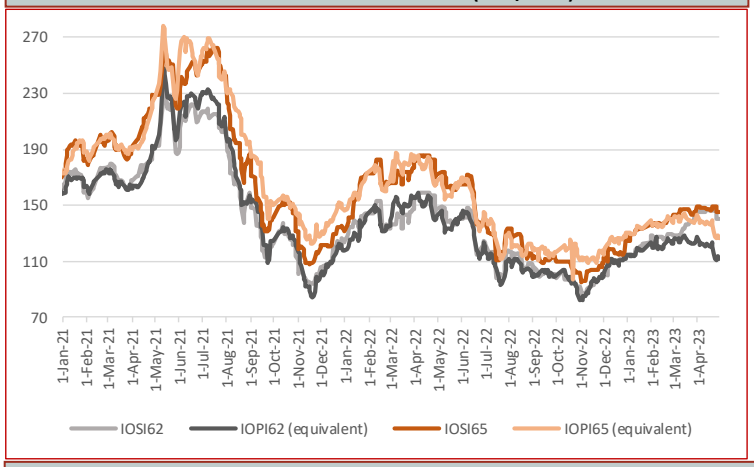
IRON ORE INDEX COMPARISONS (USD/DMT)



CHINA DOMESTIC COMPOSITE MINES INDEX (RMB/Wet Tonne, including VAT)

IRON ORE BRAND SPOT PRICE ASSESMENTS

Apr 28th, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI62
Roy Hill	772	-15	-57
SIMEC Fines	700	-16	-129
PB Fines	798	-14	-31
Newman Fines	797	-16	-32
MAC Fines	779	-15	-50
Jimblebar Blended Fines	694	-15	-135
Carajas Fines	1009	-15	180
Brazilian SSF	792	-15	-37
Brazilian Blend Fines	811	-15	-18
RTX Fines	712	-15	-117
West Pilbara Fines	740	-15	-89

Apr 28th, 2023	SEABORNE INDEX (USD/DMT)		
	Price	Change	Diff to IOSI62
Roy Hill	136.00	0.00	-4.50
SIMEC Fines	132.50	0.00	-8.00
PB Fines	136.75	0.00	-3.75
Newman Fines	139.65	0.00	-0.85
MAC Fines	136.75	0.00	-3.75
Jimblebar Blended Fines	129.15	0.00	-11.35
Carajas Fines	170.05	0.00	29.55
Brazilian SSF	144.25	0.00	3.75
Brazilian Blend Fines	145.90	0.00	5.40
RTX Fines	130.40	0.00	-10.10
West Pilbara Fines	134.75	0.00	-5.75

Apr 28th, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI58
SSF	717	-1	0
FMG Blended Fines	726	-1	9
Robe River	727	-1	10
Western Fines	729	-1	12
Atlas Fines	723	-1	6
Yandi	710	-1	-7

IRON ORE INDEX NORMALISATION DIFFERENTIALS

Port Stock Index Product Differentials (RMB/wet tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	5.00	0.00
	High Grade Fe 63 - 64%	5.00	0.00
	High Grade Fe 64 - 65%	5.00	0.00
	High Grade Fe 65 - 65.5%	5.00	0.00
	Low Grade Fe	20.00	0.00
1% Alumina	High Fe Grade Al <2.25%	30.00	0.00
	High Fe Grade Al 2.25-4%	5.00	0.00
	Low Fe Grade Al <2.25%	62.00	0.00
	Low Fe Grade Al 2.25-4%	19.00	0.00
1% Silica	High Fe Grade Si <4%	25.00	0.00
	High Fe Grade Si 4-6.5%	11.00	0.00
	Low Fe Grade	16.00	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	9.00	0.00
	High Fe Grade 0.115%<P<0.15%	9.00	0.00
	Low Fe Grade 0.09<P<0.1%	5.00	0.00

Seaborne Index Product Differentials (USD/dry tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	1.25	0.00
	High Grade Fe 63 - 64%	2.75	0.00
	High Grade Fe 64 - 65%	2.75	0.00
	High Grade Fe 65 - 65.5%	2.75	0.00
1% Alumina	High Fe Grade Al <2.25%	0.25	0.00
	High Fe Grade Al 2.25-4%	0.25	0.00
1% Silica	High Fe Grade Si <4%	1.75	0.00
	High Fe Grade Si 4 - 6.5%	4.25	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	0.50	0.00
	High Fe Grade 0.115%<P<0.15%	8.00	0.00

Port Stock Price Differentials to Qingdao Port for PB Fines (RMB/wet tonne)

Port	Value	Change	Port	Value	Change	Port	Value	Change	Port	Value	Change
Bayuquan	-100.00	0.00	Fangcheng	-25.00	0.00	Lanshan	0.00	0.00	Rizhao	0.00	0.00
Beilun	0.00	0.00	Jiangyin	-30.00	0.00	Lianyungang	0.00	0.00	Shekou	0.00	0.00
Caofeidian	-10.00	-5.00	Jingtang	-10.00	-5.00	Majishan	0.00	0.00	Taicang	-30.00	0.00
Dalian	0.00	0.00	Lanqiao	0.00	0.00	Qingdao	0.00	0.00	Tianjin	-15.00	0.00

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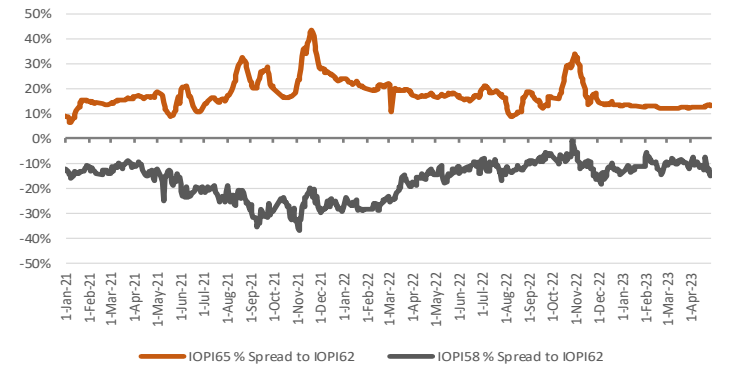
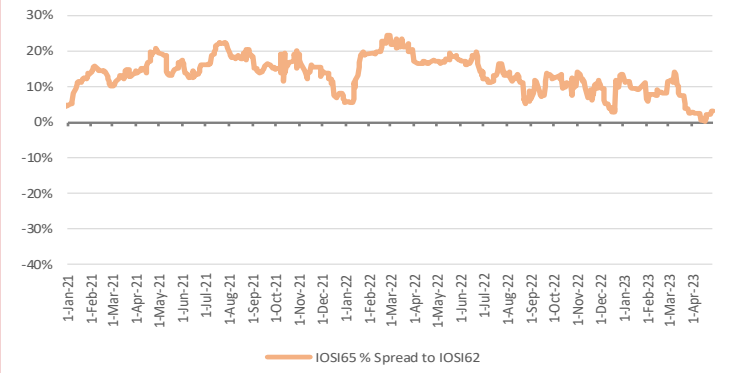
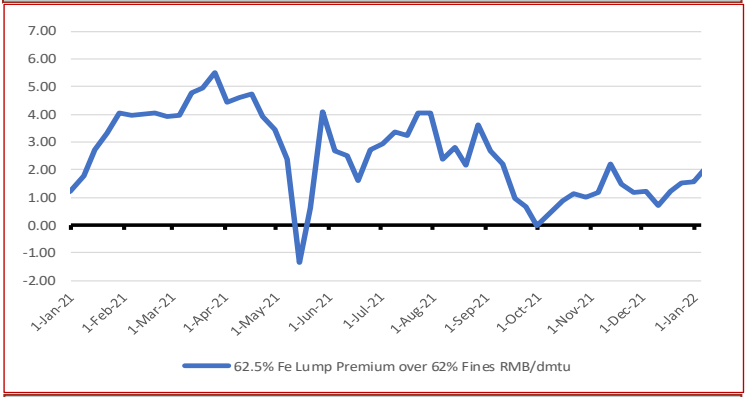
IRON ORE INDEX PREMIUMS/DISCOUNTS

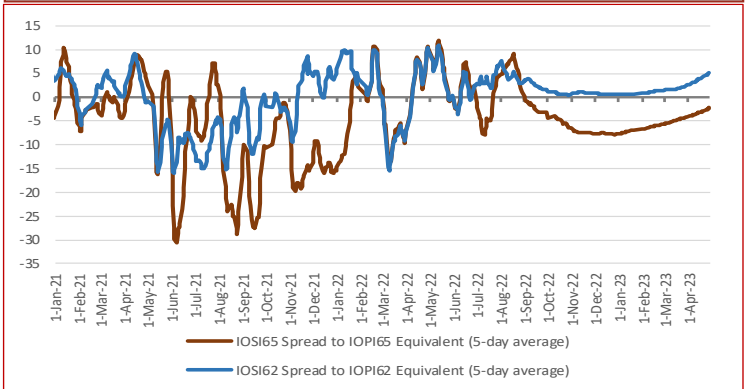




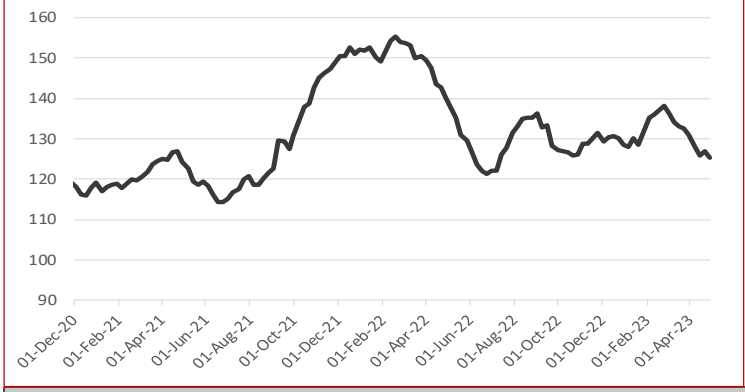
WEEKLY IRON ORE PORT STOCK LUMP PREMIUM (RMB/DMTU)



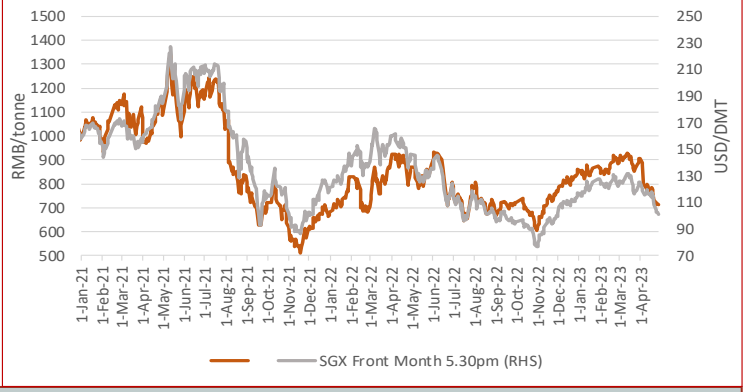
IRON ORE SEABORNE TO PORT STOCK PRICE SPREADS (USD/DMT)



TOTAL IRON ORE INVENTORIES AT CHINA PORTS (MILLION TONNES)



FUTURE TRADING—FRONT MONTH CLOSING PRICE



IRON ORE PORT INVENTORIES (MILLION TONNES)

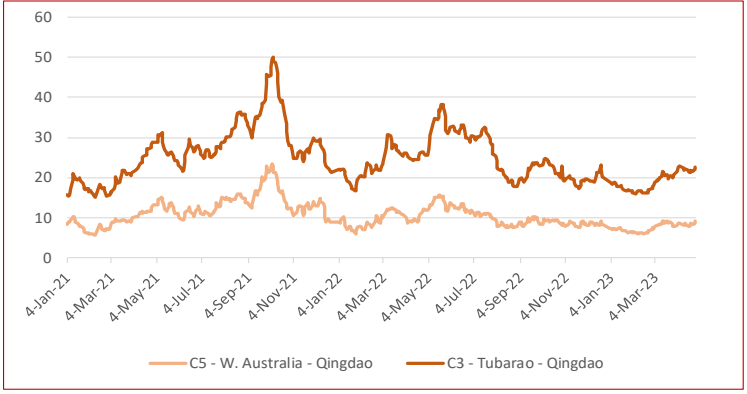
Week Ending Apr 28th, 2023

Province	This week	Change %	Low [2]	High [2]
Jingtang	12.27	0.33%	9.84	19.20
Qingdao	25.05	-0.20%	9.41	26.24
Caofeidian	12.69	-2.23%	9.05	16.29
Tianjin	9.38	1.63%	7.14	12.97
Rizhao	16.24	-3.79%	9.44	19.26
Total (35 Ports)	125.16	-1.41%	98.80	155.39

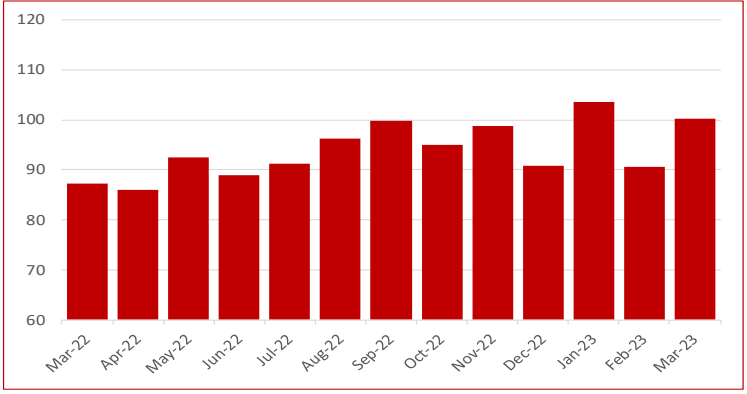
IRON ORE FUTURES CONTRACTS

	DCE (RMB/WMT)			SGX (USD/DMT)		
Closing Date	Apr 28th, 3pm close			Apr 28th, 5:30pm		
Contract	I2309	Change	Change %	May. 23	Change	Change %
Closing Price	714.00	0.50	0.07%	101.35	0.05	0.05%
Vol traded ('000 lots)	62.29	-9.13	-12.78%	3.99	-0.94	-19.14%
Open positions ('000 lots)	75.06	-1.88	-2.44%	25.42	-1.30	-4.85%
Day Low	706.0	-5.50	-0.77%	102.70	-0.95	-0.92%
Day High	723.0	-5.50	-0.75%	105.30	-0.70	-0.66%

DRY BULK FREIGHT RATES (USD/MT)



TOTAL CHINA IRON ORE IMPORT VOLUMES (MILLION TONNES)



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STEEL SPOT MARKET PRICES—CHINA

Steel Spot Market RMB/tonne

Product	2023/4/28	Change	Change %
ReBar HRB400 ϕ18mm	3,830	-140	-3.53%
Wirerod Q300 ϕ6.5mm	4,040	-140	-3.35%
HRC Q235/SS400 5.5mm*1500*C	3,980	-90	-2.21%
CRC SPCC/ST12 1.0mm*1250*2500	4,510	-100	-2.17%
Medium & Heavy Plate Q235B 20mm	4,230	-120	-2.76%
GI ST02Z 1.0mm*1000*C	4,900	-90	-1.80%
Colour Coated Plate	7,250	-150	-2.03%

CHINESE STEEL MILL PROFITABILITY

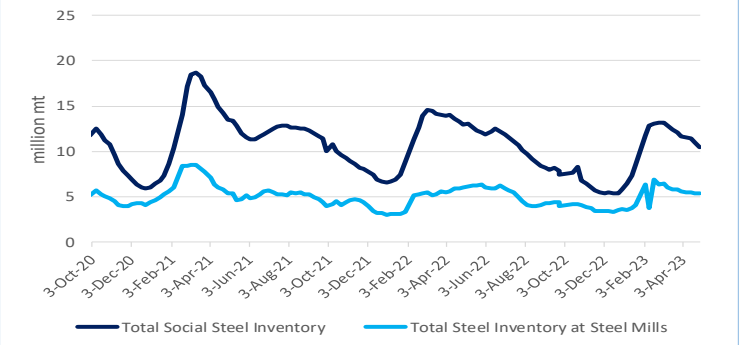
SMM Tracking of Steel Mill P&L - Rebar and Hot-rolled Coil (RMB/tonne)

Category	Price	Change (WoW)	Note
MMi (Fe 62%), USD/mt exluding tax	140.50	-5	Mmi CFR Equivalent index for 1st Feb
Coke	2,510	-200	2nd grade met coke, Tangshan, incl. tax
Steel Scrap	2,870	-150	steel scrap (6mm) in Zhangjiagang, exl. tax
Billet Cost	3,315	-163	Q234, incl. tax
Rebar cost - Blast furnace	3,779	-184	calculated based on theoretical weight, incl. tax
Rebar profit - Blast furnace	-39	14	based on Shanghai prices, incl. tax
Hot-rolled coil cost - Blast furnace	3,793	-183	based on actual weight, incl. tax
Hot-rolled coil proft - Blast furnace	167	63	based on Shanghai prices, incl. tax

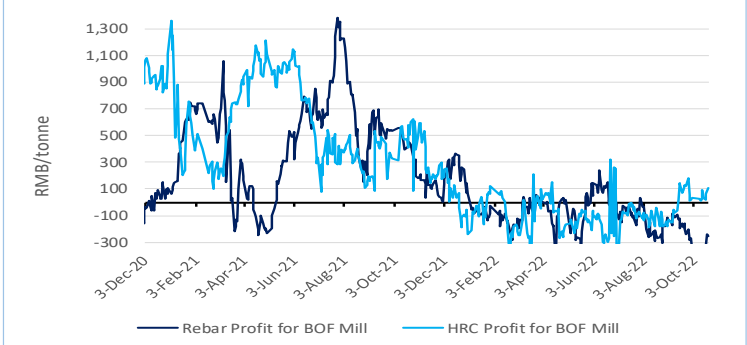
Note: 1. Costs in the table are caluclated based on todays market prices and facout our management, sales, financial and depreciations fe

2. The cost refers to average cost in the industry based on SMM's survey of small, medium and large mills in China

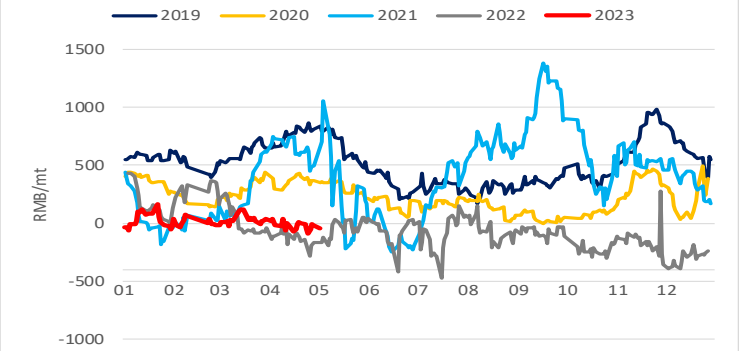
CHINESE STEEL INVENTORIES



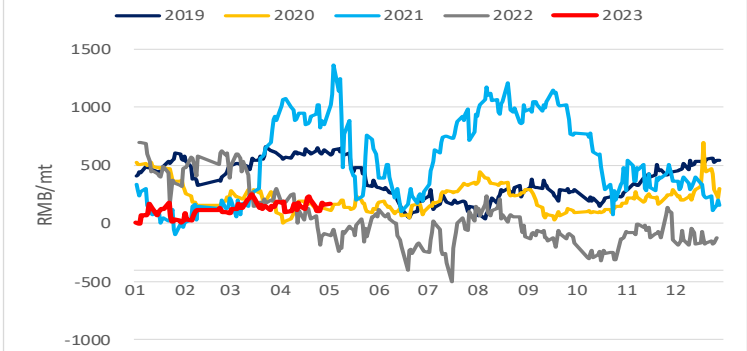
CHINESE STEEL MILL PROFITABILITY



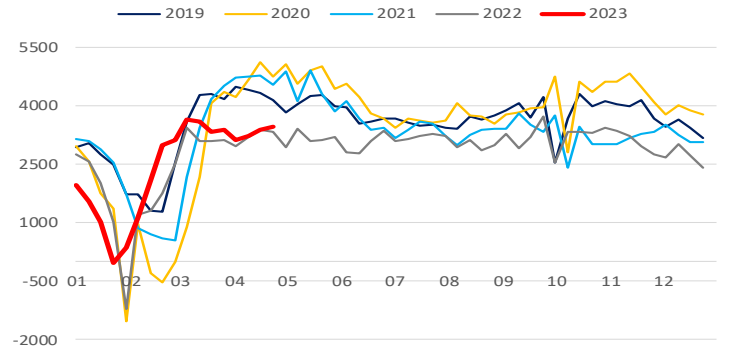
CHINESE STEEL MILL PROFITABILITY—Rebar



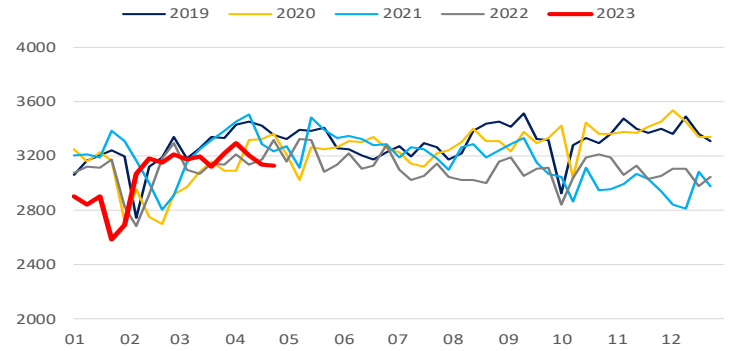
CHINESE STEEL MILL PROFITABILITY—Hot-rolled Coil



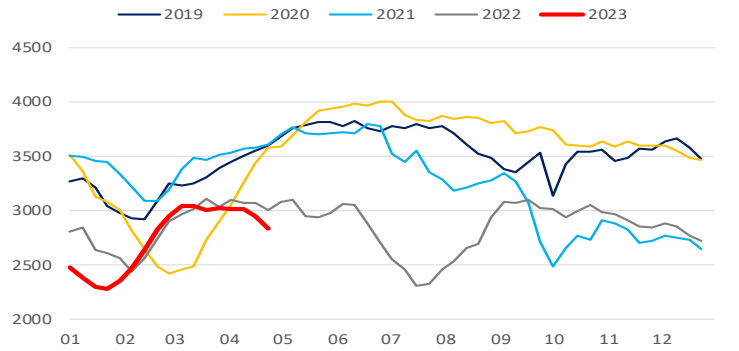
CHINESE STEEL CONSUMPTION—Rebar



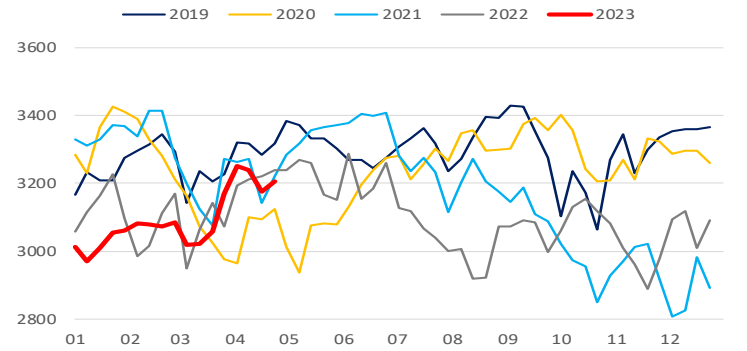
CHINESE STEEL CONSUMPTION—Hot-rolled Coil



CHINESE STEEL PRODUCTION—Rebar



CHINESE STEEL PRODUCTION—Hot-rolled Coil



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IRON ORE INDEX SPECIFICATIONS, COMPILATION RATIONALE AND DATA EXCLUSIONS

Iron Ore Index Specifications (Port and Seaborne)

	65% Fe Fines	62% Fe Fines	58% Fe Fines	62.5% Fe Lump
Fe %	65.00	62.00	58.00	62.50
Alumina %	1.40	2.25	2.25	1.50
Silica%	1.50	4.00	5.50	3.50
Phosphorus %	0.06	0.09	0.05	0.08
Sulphur %	0.01	0.02	0.02	0.02
Moisture %	8.00	8.00	9.00	4.00
Sizing	Granular size below 10mm for at least 90% of cargo; maximum of 40% below 150 micron			Size below 6.3mm max 15% Size above 31.15mm max 25%
Pricing Point	Qingdao Port (FOT and CFR respectively)			FOT Qingdao Port
Timing (Seaborne)	Loading within 4 weeks, Delivery within 8 weeks			Delivery within 2 weeks
Payment Terms	L/C at sight			L/C at sight or CAD

Iron Ore Index Compilation Rationale and Data Exclusions

MMi iron ore indices are compiled from data provided by companies that are part of the iron ore supply chain and involved in spot market transactions. The indices are calculated using detailed transaction-level data submitted to MMi by these companies. This data is normalised to the appropriate specifications and screened to remove outliers before volume-weighted average prices are calculated from the remaining core set of data.

For more details on MMi's iron ore methodology please download the guide published on our website at: www.mmiprices.com

Data Exclusions*

	62%	58%	65%
Port Index	0	0	0
Seaborne index	0		0
Lump Index 62.5	0		

* Number of price submissions for iron ore indices that were excluded from index calculations today as they were anomalous and could not be verified

IRON ORE DOMESTIC CONCENTRATE INDEX CALCULATION METHODOLOGY

The compilation method for price index generally refers to the compilation method of CPI price index and other price indices, breakdown the price data and calculate the average value according to a certain method, taking the vertical axis as the regional composite index (average of different grade index) and the horizontal axis as the grade composite index (average of different regional index), a total composite index for domestic ore can be output ultimately. The process system is also adopted in the calculation i.e. each sub-index can be obtained as well.

AVERAGE IRON ORE SPECIFICATIONS APPLIED FOR BRAND PRICE ASSESSMENTS

PORT STOCK BRANDS

Apr 28th, 2023	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.88%	2.35%	4.45%	0.055%	9.26%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines	61.64%	2.31%	3.87%	0.099%	9.45%
Newman Fines	62.49%	2.31%	3.98%	0.090%	7.72%
MAC Fines	60.43%	2.38%	4.64%	0.078%	7.57%
Jimblebar Blended Fines	60.87%	2.95%	4.53%	0.111%	7.32%
Carajas Fines	64.90%	1.39%	1.82%	0.080%	7.78%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	62.56%	1.58%	4.79%	0.091%	8.72%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

Apr 28th, 2023	Specifications applied for 58% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
SSF	56.49%	3.20%	6.19%	0.065%	9.18%
FMG Blended Fines	58.20%	2.43%	5.54%	0.057%	8.29%
Robe River	56.44%	3.16%	5.73%	0.042%	8.44%
Western Fines	57.88%	2.87%	7.50%	0.062%	7.45%
Atlas Fines	58.00%	1.85%	5.50%	0.090%	9.00%
Yandi	56.87%	1.58%	6.41%	0.042%	9.53%

SEABORNE BRANDS

	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.70%	2.30%	4.90%	0.055%	8.00%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines 62%	62.00%	2.60%	4.30%	0.090%	9.50%
Newman Fines	62.30%	2.40%	4.30%	0.095%	8.00%
MAC Fines	60.80%	2.35%	4.40%	0.075%	7.80%
Jimblebar Blended Fines	60.50%	3.10%	4.80%	0.120%	7.80%
Carajas Fines	65.10%	1.50%	1.70%	0.080%	8.50%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	63.00%	1.50%	5.00%	0.070%	7.00%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

BLOOMBERG TICKERS

PORT STOCK INDICES

	FOT Qingdao (RMB/wet tonne)	CFR Qingdao Equivalent (USD/dry tonne)
IOPI62	IRCNQ001	IRCNQ004
IOPI58	IRCNQ002	IRCNQ005
IOPI65	IRCNQ003	IRCNQ006
IOPLI62	IRCN0036	IRCN0037

SEABORNE INDICES

	CFR Qingdao (USD/DMT)
IOSI62	IRCN0034
IOSI65	IRCN0035

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